

SARASWAT ACADEMY

Chapter Test – Mathematics

Class IX | Chapter 7: The Mathematics of Maybe: Introduction to Probability

1. A fair six-sided die is rolled once. What is the probability of getting a number greater than 4? [1]
(a) $1/6$ (b) $1/3$ (c) $1/2$ (d) $2/3$
2. When two fair coins are tossed simultaneously, the sample space is [1]
(a) {H, T} (b) {HH, HT, TH, TT} (c) {HH, TT} (d) {HHT, TTH}
3. If an experiment is performed 50 times and an event occurs 15 times, its experimental probability is [1]
(a) $3/10$ (b) $7/10$ (c) $15/50$ (d) Both (a) and (c)
4. Which of the following events has probability 0? [1]
(a) Getting a number less than 7 on a standard die
(b) Getting an even number on a standard die
(c) Getting 8 on a standard die
(d) Getting a number from 1 to 6 on a standard die

SECTION B – SHORT ANSWER QUESTIONS

5. A fair die is rolled once. Write the sample space and find the probability of getting an odd number. [2]
6. Two fair coins are tossed together. Find the probability of getting exactly one head. Show the favourable outcomes. [2]
7. A die is rolled 60 times and the number 5 occurs 12 times. Find the experimental probability of getting 5. Also state the theoretical probability of getting 5 on a fair die. [2]

SECTION C – LONG ANSWER QUESTIONS

8. A box contains 5 green balls and 7 red balls. One ball is drawn at random.
(i) Write the sample space in terms of the possible colours.
(ii) Find the probability of drawing a green ball.
(iii) Find the probability of drawing a red ball. [3]
9. A child has 2 shirts—one red and one blue—and 3 types of pants—jeans, khakis and shorts.
(i) List all possible shirt–pants combinations.
(ii) How many outcomes are possible?
(iii) If one combination is selected at random, what is the probability that the shirt is red? [3]

SECTION D – CASE STUDY

10. Case Study: A Game of Chance

At a school mathematics fair, a student tosses a fair coin twice. Each complete result is recorded. The teacher explains that the list of all possible outcomes is called the sample space. The student then studies the event of getting two heads.

- a. Write the sample space for the experiment. [1]
- b. How many outcomes are there in the sample space? [1]
- c. Write the event corresponding to getting two heads. [1]
- d. Find the probability of getting two heads. [1]